

QUANTUM COMPUTING: A DOUBLE-EDGED SWORD FOR DIGITAL BANKING SECURITY

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INTRODUCTION:

- As we all are aware that quantum computing is a novel approach to computation that leverages the principles of quantum mechanics to solve complex problems far beyond the capabilities of classical computers.
- Unlike classical computers, quantum computers use the principles of superposition and entanglement to solve complex problems exponentially faster, enabling enhanced fraud detection, sophisticated risk forecasting, and real-time analysis of massive financial datasets.
- As quantum computing is rapidly transforming the landscape of the digital world it is now becoming a threat towards digital banking and financial security, presenting both unprecedented opportunities and significant risks to existing systems.
- This immense computing power is also threatening traditional cryptographic standards upon which current financial transactions and digital banking systems are built. Algorithms such as RSA(Rivest-Shamir-Adleman) which relies its security on the mathematical difficulty of factoring large prime numbers and ECC(Elliptic Curve Cryptography) which is a public-key cryptography method based on the algebraic structure of elliptic curves over finite fields, providing efficient and secure encryption for data with smaller key sizes, are long used for securing sensitive banking information, could be rendered obsolete by quantum attacks whereby hostile actors could break encryption and even harvest encrypted data now for future decryption, a threat known as "harvest now, decrypt later" (HNDL).
- To meet the urgency of this transition, banks and financial institutions are investing in post-quantum cryptography and quantum key distribution to secure their infrastructures and maintain global financial stability. The coming era of quantum

technologies demands cross-industry collaboration and immediate action to build quantum-safe security frameworks that are resilient to future threats.

PROPOSED SOLUTION:

- The proposed solution to secure digital banking in the quantum era is a comprehensive, multi-layered strategy combining post-quantum cryptography (PQC), quantum key distribution (QKD), cryptographic agility, and hybrid security models.
- Some of these solutions in detail are:

➤ Post-Quantum Cryptography (PQC):

Implement new cryptographic algorithms, standardized by organizations such as NIST, that are resistant to quantum attacks and designed to replace vulnerable public-key cryptosystems like RSA and ECC.

Begin the transition now to ensure all critical systems—payment networks, SWIFT messaging, and blockchain-based transactions—can operate with PQC.

➤ Quantum Key Distribution (QKD):

Deploy QKD for ultra-sensitive data links where the highest level of security is required; QKD uses quantum mechanics to guarantee that any interception of cryptographic keys will be detected.

Examples include pilot QKD-secured network links developed by HSBC and JP Morgan Chase.

➤ Crypto-Agile & Hybrid Models:

Upgrade infrastructure to support cryptographic agility, allowing systems to quickly swap between traditional and quantum-safe algorithms as standards evolve.

Adopt hybrid approaches during the transition phase: run classical and quantum-safe encryption side by side, minimizing risk if a vulnerability is discovered in either.

➤ Cryptographic Asset Inventory & Phased Migration:

Maintain an inventory of all cryptographic assets, including algorithms, keys, and business functions, to plan and prioritize upgrades.

Coordinate planning and roll out phased migration of quantum-safe solutions across the organization, integrating compliance, incident response, and key management with new regulatory requirements.

TECHNICAL WORKFLOW:

Some Simple Steps for Quantum-Safe Security that we can use are:

- **Make a list:** Write down all types of data and security methods used in the bank systems. This helps to find out which parts are the most sensitive and need the best protection.
- **Check for risks:** See which security methods (like RSA keys or certificates) could become weak if quantum computers are used by hackers. Focus first on protecting important data that needs to stay safe for years.
- **Make a plan:** Decide which new quantum-safe methods should be used, how to update or replace old systems, and when each upgrade should happen. Work with technology partners to avoid mistakes and keep things running smoothly.
- **Put new protections in place:** Carefully swap out the vulnerable security methods for stronger, quantum-safe versions, like PQC (Post-Quantum Cryptography) and, for very sensitive uses, Quantum Key Distribution (QKD).
- **Test and watch:** Use tests to make sure the new protections work as expected. Set up ongoing monitoring during and after the changes to make sure security stays strong and nothing slips through.
- **Stay up to date:** Regularly review the system, update protections, and learn about new threats and standards, so the bank always stays ahead.

RECENT ACTIVITIES RELATED:

- A leading example is the successful rollout of Quantum Key Distribution (QKD) by a major UAE bank working with a QKD technology provider, aiming to protect its transaction and customer data against cyber threats. This bank started with pilot projects, testing QKD in real conditions and then gradually adopted it across sensitive channels. The project involved strategic consulting, leader training, and careful integration, resulting in stronger data protection and recognition as a cybersecurity innovator.
- We could also see this in Switzerland and Austria where financial institutions employ QKD for data center backup transmissions, protecting long-term banking records.

- Even British Telecom (UK) offers QKD as a service over fiber links, helping customers secure critical data streams.

CONCLUSION:

- In conclusion, the rise of quantum computing represents a transformative challenge and opportunity for digital banking security. Quantum computers have the power to break current cryptographic methods, putting sensitive financial data at risk of future decryption by malicious actors. This urgency demands banks and financial institutions to adopt quantum-safe technologies without delay. As we have seen a few real-world examples that we have been observing till now and this trend is set to go further with emergence of quantum computing in maybe 5 to 10 years and we will have to adapt these new techniques in the near future.

REFERENCES:

- How does a bank become quantum cybersecure? – Luxoft
- Securing information in the age of quantum cyberspace-bankingjournal.aba
- Quantum-readiness for the financial system: a roadmap-bis.org
- Quantum Internet: Quantum Key Distribution in Banking-infotech.co